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Lab 21

Part A

A1

```
SELECT TOP 1 Id, FullName, YearOfStudy, Gpa
FROM dbo.Students
ORDER BY (Id % 27), Id;
```

8 %

Results Messages

	Id	FullName	YearOfStudy	Gpa
1	1	Kareem Fouad	4	3.2

A2

```
--- AGENT (picks a tool, or falls back to RAG) ---
Q: What honour band is Kareem Fouad in?
[tool used: student_lookup]
A: Kareem Fouad is a Year 4 student with a GPA of 3.20, which places them in the Second honour band.

This is the day's payoff: the same question the Warm-Up predicted
on paper, now answered correctly " because the agent was given the
one thing a bare model never had: your own real data.
PM>
```

A4

```
one thing a bare model never had: your own real data.
AGENT instructor_lookup
Q: Who teaches Web Development Using .NET?
tool used: instructor_lookup
A: Hamdy teaches Web Development Using .NET.
PM> |
```

Part B

B1

If my agent's `InstructorLookupTool` could WRITE to the database instead of only reading from it say an `AssignInstructorTool` that updates which instructor is assigned to a course the one guardrail I'd add first is never let it perform a direct `UPDATE` on `dbo.Courses`. Instead, it should create a pending `InstructorChangeRequest` record that a human has to approve before the real assignment changes

B2

The `StudentPortalWeb` ASP.NET Core MVC/Razor Pages site, with conventional and attribute routing (Lab 16)

the `StudentsController` `Edit` action pair, with the guard - clause - before - save pattern and a proper redirect (not a view return) on a successful POST (Lab 17)

the write side of `Enrollment`: a full `EnrollmentsController` built from scratch (GET + POST) (Lab 19)

`StudentPortalAI` — the RAG pipeline and agent built in Session 21, including my own `InstructorLookupTool`, all answering real questions grounded in the real `ITI_StudentPortal` data